ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PRACTICE EXAM 1

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 8 year: bridges and frame structures, forces, struts and ties, stability, plastics and other materials, gears, pulleys and belts, mechanical advantage, AND and OR logic circuits, LEDs, and orthographic drawing.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings and tables.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 A bridge that is made strong by a framework of triangles is a …

A beam bridge. B arch bridge. C truss bridge. D pontoon bridge.

1.2 A member of a structure that is in compression is called a …

A strut. B tie. C cable. D gusset.

1.3 A gear placed between the driver and driven gears so that they turn in the same direction is called …

A a bevel gear. B an idler gear. C a worm gear. D a rack.

1.4 A driver gear with 20 teeth turns a driven gear with 40 teeth. The driven gear turns …

A twice as fast as the driver. B at the same speed.
C four times faster. D at half the speed of the driver.

1.5 A rack and pinion changes …

A rotary motion into linear motion. B linear motion into sound.
C slow rotation into fast rotation only. D electrical energy into motion.

1.6 Two switches connected in series to control one lamp work like …

A an OR gate. B a NOT gate. C an AND gate. D a fuse.

1.7 A crossed belt between two pulleys makes the driven pulley turn …

A in the same direction as the driver. B faster in every case.
C in the opposite direction to the driver. D not at all.

1.8 Which of these is a thermoplastic?

A Acrylic (Perspex) B Bakelite C Melamine D Epoxy resin

1.9 In first-angle orthographic projection, the top view (plan) is drawn …

A above the front view. B to the left of the front view. C on a separate page. D below the front view.

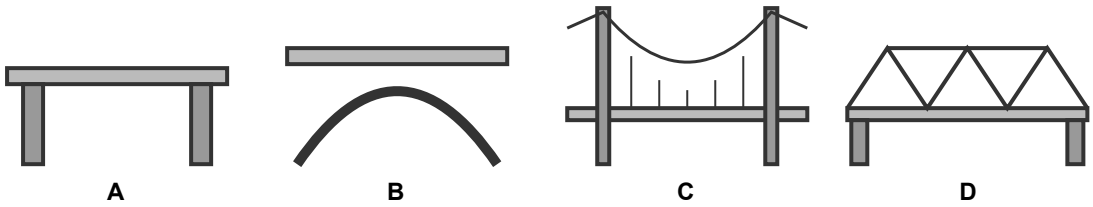
1.10 The SI unit of force is the …

A kilogram. B newton. C pascal. D joule.

[10]

QUESTION 2: STRUCTURES

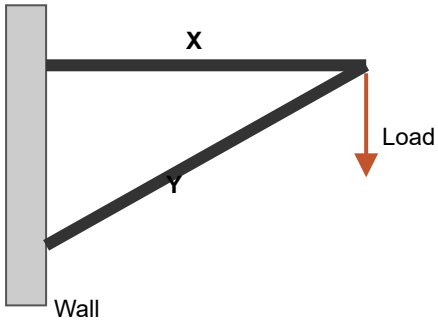
2.1 Identify the types of bridges labelled A, B, C and D.



(4)

2.2 Which type of bridge is best for crossing a very wide, deep river? Give a reason. (2)

2.3 A shelf bracket is fixed to a wall. A heavy load hangs from the outer end.



2.3.1 Is member X a strut or a tie? (1)

2.3.2 Is member Y a strut or a tie? (1)

2.3.3 Explain your answer to 2.3.2. (1)

2.4 Name the force acting in each case: (a) a plank that sags when a person stands in the middle; (b) scissors cutting paper; (c) a screwdriver turning a tight screw. (3)

2.5 What is the centre of gravity of an object? Give TWO ways to make a structure more stable. (3)

2.6 Explain why steel bars are placed inside concrete beams. (3)

[18]

QUESTION 3: MATERIALS AND PROCESSING

3.1 Explain the difference between thermoplastics and thermosetting plastics, and give ONE example of each. (4)

3.2 Name ONE industrial process used to shape plastic, and name a product made by this process. (2)

3.3 Plastic bottles are marked with a recycling code inside a triangle, e.g. 1 = PET. Why are these codes useful? (2)

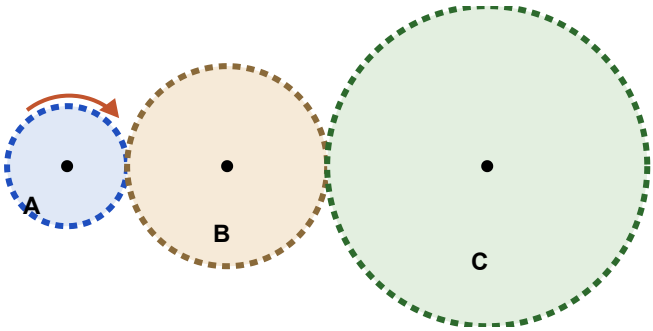
3.4 Explain the difference between ferrous and non-ferrous metals. Give ONE example of each. (2)

3.5 Give TWO safety precautions when heating or cutting plastic in the workshop. (2)

[12]

QUESTION 4: MECHANICAL SYSTEMS

4.1 Study the gear train. Gear A is the driver.



Gear A has 15 teeth, gear B has 25 teeth and gear C has 45 teeth. Gear A turns clockwise.

4.1.1 In which direction does gear C turn? (1)

4.1.2 Calculate the velocity ratio between gear A and gear C. (2)

4.1.3 Gear A turns at 90 r/min. Calculate the speed of gear C. (2)

4.1.4 What is the purpose of gear B? (1)

4.2 A driver pulley with a diameter of 40 mm turns at 1 200 r/min. It drives a pulley with a diameter of 160 mm. (a) Calculate the speed of the driven pulley. (b) Write down the velocity ratio. (c) How can the driven pulley be made to turn in the opposite direction? (4)

4.3 Give ONE use of each: (a) a worm and wheel; (b) bevel gears; (c) a rack and pinion. (3)

4.4 A pulley system with four supporting ropes is used to lift a load of 800 N. Ignoring friction, calculate the effort needed and the mechanical advantage. (2)

4.5 Explain why a cyclist changes to a low gear (large rear sprocket) when riding up a steep hill. (3)

[18]

QUESTION 5: ELECTRICAL SYSTEMS

5.1 Two switches, A and B, are connected in series with a lamp and a battery. Copy and complete the truth table (0 = off/open, 1 = on/closed). Name the type of logic.

Switch A	Switch B	Lamp
0	0	
0	1	
1	0	
1	1	

(4)

5.2 (a) Two switches are now connected in parallel to control one lamp. Name the type of logic. (b) Give ONE practical use of two switches in parallel. (3)

5.3 Give ONE example of a machine where AND logic is used for safety, and explain why. (2)

5.4 (a) Why must a resistor be connected in series with an LED? (b) What happens if an LED is connected the wrong way round? (c) Give ONE advantage of an LED over a filament bulb. (3)

5.5 Two identical bulbs are connected (i) in series and (ii) in parallel to the same battery. (a) In which circuit are the bulbs brighter? (b) What happens to the other bulb in each circuit if one bulb blows? (4)

5.6 Name the safety device that melts and breaks a circuit if the current becomes too high. (1)

[17]

TOTAL: 75

END OF QUESTION PAPER